

VIThelpdesk

AI Assistant with Intelligent Tool Selection

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Project Objectives

- Automate college student services.
- Detect precise user intent.
- Execute correct backend tools.
- Deliver accurate responses fast.

Working Methodology

- Receive student request query.
- Apply rule-based intent detection.
- Route to specific tool.
- Retrieve local RAG FAQs.
- Return clear meaningful response.

System Architecture

- Frontend: HTML, CSS, JavaScript
- Backend: FastAPI (Python)
- Intent Engine: Rule-Based Logic
- Knowledge Base: JSON + RAG
- Database: SQLite

Approach Comparison

- Rule-Based: Fast, local, zero cost.
- LLM-Based: Semantic, slow, expensive API.
- Hybrid: Balanced, higher architectural complexity.
- Selected: Rule-Based for reliability.
- Avoids external API rate limits.

Evaluation Results

- Clear requests tested
- Ambiguous requests tested
- Multi-step requests tested

Evaluation Screenshot

```
(venv) sravanthi-murukuri@sravanthi-murukuri-HP-Laptop-15s-fr4xxx:~/ai-support-chatbot$ /home/sravanthi-murukuri/ai-support-chatbot/venv/bin/python /home/sravanthi-murukuri/ai-support-chatbot/test_eval.py
[✓] FAISS Vector Database loaded into memory successfully!
=====
🔥 RUNNING UPGRADED CHATBOT EVALUATION
=====

=====
📊 EVALUATION RESULTS SUMMARY
=====
• Total Test Cases Evaluated      : 53
• Intent Detection Accuracy       : 73.6% (39/53)
• Tool Selection Accuracy        : 96.0%
• Response Quality / Success Rate: 100.0% (53/53)
=====
[✓] Upgraded evaluation completed successfully!
(venv) sravanthi-murukuri@sravanthi-murukuri-HP-Laptop-15s-fr4xxx:~/ai-support-chatbot$
```

Challenges Faced

- Managing external API limits.
- Resolving UI theme layout bugs.
- Fixing inconsistent query variations.
- State synchronization across steps.

Future Improvements

- Add an admin management dashboard.
- Integrate dynamic external APIs.
- Expand local semantic capabilities.
- Multi-lingual support.

Thank You

Questions?